

# Autodesk® Scaleform®

## Scaleform 4.4

Scaleform 4.4

: Matthew Doyle  
: 2.10  
: 2013 4 24

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## Autodesk® Scaleform® 4.4

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Scaleform 4.3

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# 1

Autodesk® Scaleform® 4.4 . Scaleform® 4.4 ,  
SDK ,  
 . Scaleform 4.4 , ( )  
 . , Scaleform Common Lightweight Interface Kit (CLIK™) Video  
 .

Scaleform 4.4 :

- x Scaleform CLIK –
- x Scaleform Analyzer for Memory and Performance (AMP™) –
- x ActionScript 2 ActionScript 3
- x , 2.5D
- x Scaleform Kits - UI ( )
- x GfxExport –
- x
- x CRI Movie™ Scaleform Video
- x Scaleform Input Method Editor (IME)

## 2

Scaleform 4.4 SDK    Scaleform Video, Scaleform IME

<http://gameware.autodesk.com/scaleform/developer/?action=dl>

### 2.1 *Flash*

Scaleform 4.4    Adobe® Flash® Creative Suite® 3 (CS3)    . Scaleform 4.4    Flash  
(AS)    . Scaleform 4.4    Flash Player 10.1

. Scaleform 4.4    AS2.0    AS3.0    . Scaleform 4.4    Flash  
2.0    *Scaleform 4.4– Flash Support Overview*

Creative Suite(CS)

CS

CS4

CS3

Flash, ,

### 2.2 *SDK*

Scaleform 4.4 SDK

*C:\Program Files (x86)\Scaleform\GFx SDK*

4.4

Scaleform 4.4



Scaleform 4.4 SDK

SDK

readme\_make.txt

readme\_xx.txt

DirectX® SDK

, DirectX SDK Scaleform 4.4

Player

Scaleform 4.4 Player DirectX

SDK

Scaleform →

GFx SDK 4.4 → Documentation

*C:\Program Files (x86)\Scaleform\GFx SDK 4.4\Doc*

## 2.2.1

Scaleform

1.

2.

3.

Scaleform

SDK

Scaleform IME

Scaleform

(

).

Scaleform

(

)

.

SDK

GFXSDK

. Scaleform

.tar.gz

.zip

Scaleform

.

Scaleform

make

. Scaleform SDK

FxPlayer, PlayerSWFToTexture

( )

,

Visual Studio

|                  |             |
|------------------|-------------|
|                  |             |
| GFX_ENABLE_VIDEO | Libgfxvideo |
| GFX_ENABLE_IME   | Libgfxime   |

Scaleform

Shipping

Release

Debug

DebugOpt

### 2.2.1.1 Windows OS

tar.bz2

Windows

7-zip

tar.bz2

## 2.2.2 SDK

Scaleform 4.4 SDK ,  
.  
SDK ( -> Scaleform -> GfX 4.4->Scaleform SDK  
Browser ) .  
scaleform\_gfx\_4.4\_macos/Bin/Browser/ Browser.app SDK Browser  
.

## 2.2.3 Scaleform CLIK

Scaleform 4.4 Scaleform Common Lightweight Interface Kit (CLIK) , CLIK  
(UI) . SDK

CLIK Flash  
. CLIK *Getting Started with CLIK* .

1.

→ → Scaleform → GfX SDK 4.4 → Documentation → Scaleform 4.4 – Getting Started  
with CLIK.pdf

2.

C:/Program Files/Scaleform/GfX SDK 4.4/Doc/sf\_4.4\_getting\_started\_with\_CLIK.pdf

3. SDK

CLIK Scaleform SDK .

:

*C:/Program Files/Scaleform/GfX SDK 4.4/Resources/AS2/CLIK/*

*C:/Program Files/Scaleform/GfX SDK 4.4/Resources/AS3/CLIK/*

*C:/Program Files/Scaleform/GfX SDK 4.4/Resources/Tools/*

:

scaleform\_gfx\_4.4\_platform/*Resources/AS2/CLIK/*

scaleform\_gfx\_4.4\_platform/*Resources/AS3/CLIK/*

scaleform\_gfx\_4.4\_platform/*Resources/Tools/*

## 2.2.4 Scaleform Video

CRI Movie                      Scaleform Video                      (Scaleform                      )

Scaleform 4.4 SDK                      . Scaleform                      Flash

UI      Flash                      .

Scaleform                      "PlayStation® 3"(PS3™), Xbox® 360, Wii™, Mac      PC                      .

Scaleform Video                      Scaleform → GfX SDK 4.4 → Video

. Scaleform Video                      *Getting Started with Video*

.

Video                      : *C:/Program Files/Scaleform/GfX SDK 4.4/Bin/Tools/VideoEncoder/*

Video                      : *C:/Program Files/Scaleform/GfX SDK 4.4/Bin/Data/AS2/Video/*

Video                      :

1. *C:/Program Files/Scaleform/GfX SDK 4.4/Doc/sf\_4.4\_getting\_started\_with\_video.pdf*

2.                      →                      → Scaleform → GfX SDK 4.4 → Documentation → Scaleform 4.4 – Getting Started with Video.pdf

3. SDK Browser.

## 2.2.5 Scaleform IME

Scaleform                      (IME)      Flash                      .

PC                      .

Scaleform 4.4 SDK

*C:/Program Files/Scaleform/GFx SDK 4.4/Bin/Data/AS2/IME/*

*C:/Program Files/Scaleform/GFx SDK 4.4/Bin/Data/AS3/IME/*

## 2.2.6 Platform SDKs

Scaleform

SDK

Scaleform

SDK

Scaleform

[Build](#)

[History](#)

## 2.2.7 .Net

Scaleform Video CRI movie

*NET Framework 2.0 SP1*

[Microsoft .NET Framework Version 2.0 SP1](#)

## 2.2.8

Scaleform

Scaleform

3rdParty/

Apps/

Samples

Common

FxPlayer

Scaleform Flash player.

GFxPlayerTiny

Scaleform

Bin/

( , )

|                  |                                 |                          |
|------------------|---------------------------------|--------------------------|
| [Platform]       | (ScaleformPlayer,               | )                        |
| Data/AS2/Samples | .                               |                          |
| FxPlayer         | ScaleformPlayer                 |                          |
| RenderTexture    | .                               |                          |
| SWFToTexture     | .                               |                          |
| ImageDelegate    | ImageDelegate                   | Data/AS2/Video Scaleform |
| .                |                                 |                          |
| Data/AS2/IME     | Scaleform IME sample files      | Data/AS2/Kits 2          |
| Data/AS3/Samples | 3                               |                          |
| Data/AS3/IME     | Sample IME ActionScript 3 files |                          |
| Data/AS3/Kits    | 3                               |                          |
| Doc/             |                                 |                          |
| Include/         | Scaleform ( C++ )               |                          |
| Lib/             | Scaleform                       |                          |
| [Platform]       |                                 |                          |
| [Config]         | .                               |                          |
| Obj/             | .                               |                          |
| [Platform]       | .                               |                          |
| Projects/        | .( , .)                         |                          |
| Common           | .                               |                          |
| [Platform]       | .                               |                          |
| Src/             | Scaleform                       |                          |
| GFx              | Scaleform SDK                   |                          |
| Render           | ( .)                            |                          |
| Sound            | ( .)                            |                          |

Scaleform

#define

| Scaleform |                       | #defines                   |
|-----------|-----------------------|----------------------------|
| Win32     | Windows               | SF_OS_WIN32                |
| x64       | Windows for x86_64    | SF_OS_WIN32                |
| *-linux   | Linux for * processor | SF_OS_LINUX                |
| MacOS     | MacOS X               | SF_OS_DARWIN,<br>SF_OS_MAC |
| Android   | Android               | SF_OS_ANDROID              |
| iPhone    | iPhone                | SF_OS_IPHONE               |
| PS3       | Playstation 3         | SF_OS_PS3                  |
| NGP       | NGP                   | SF_OS_NGP                  |
| 3DS       | 3DS                   | SF_OS_3DS                  |
| Wii       | Wii                   | SF_OS_WII                  |
| Xbox360   | Xbox 360              | SF_OS_XBOX360              |

|        | Standard, often gcc       |
|--------|---------------------------|
| Msvc80 | Visual Studio 8.0 (2005)  |
| Msvc90 | Visual Studio 9.0 (2008)  |
| Msvc10 | Visual Studio 10.0 (2010) |
| CW     | Codewarrior               |
| snc    | SN Systems                |

[Platform]

. Win32/Msvc80

. "standard"

MacOS  
 . "standard"

## 2.2.9 Scaleform

Scaleform Visual Studio Visual Studio  
 IDE make 'Projects'

### 2.2.9.1 make

Visual Studio IDE 'make' . Scaleform  
 SDK make *readme\_make.txt*

- x Linux
- x MacOS
- x Android™
- x iPhone®
- x PS3
- x Nintendo 3DS™
- x Wii
- x Wii U™

make Scaleform make Linux  
**cygwin**  
 Windows "Cygwin Bash Shell" "ld" "make"

1. setup.exe . <http://www.cygwin.com/>
2. 'Install from Internet' .
3. . (c:/cygwin)



4. . (c:/cygwin)

5. 'Internet Connection' 'Direct Connection' .

6. , .

7. 'Select Packages' 'Devel' , 'binutils' 'make' .

8. .

, "Cygwin Bash Shell" "ld" "make" .

```
$ make -v
```

```
GNU Make 3.81
```

```
Copyright (C) 2006 Free Software Foundation, Inc.
```

```
...
```

```
This program built for i686-pc-cygwin
```

```
$ ld -v
```

```
GNU ld (GNU Binutils) 2.18.50.20080625
```

```
===
```

Scaleform .

make .

1. Scaleform Makeconfig .(  
.) .

a. export CELL\_SDK := <path to cell sdk>

b. EPATH := \$(PATH)

c. export PATH = \$(CELL\_SDK)/host-win32/ppu/bin:\$(CELL\_SDK)/host-  
i. win32/Cg/bin:\$(CELL\_SDK)/host-win32/bin:\$(EPATH)

2. Makeconfig

. Use /<drive letter>/... (msys ) /cygdrive/<drive  
letter>/... (Cygwin ) , CELL\_SDK C:\ps3\cell

a. "export CELL\_SDK := /cygdrive/c/ps3/cell".

3.

a. \$ make P=PS3

b. \$ make P=PS3 C=Debug

```
make          C=<config>          .          "+"
          .          , RTTI          "make C=Release+NoRTTI"
          .          Makefile          ,          Projects
          makefile          .
```

### 2.2.9.2 'make' :

```
          :
make P=Android C=Debug
make P=3DS C=Debug+NoRTTI
```

**SNC** **PS3**  
make P=PS3+snc

**GCC** **PS3**  
make P=PS3

**non-SCU lumping** **iOS** **(SCU** **)**  
make P=iPhone/armv7 C=Release

```
          :
make P=wii C=Shipping+NoRTTI VERBOSE=1
```

**Mac** **RTTI** **x64** **:**  
make P=MacOS/x86\_64 C=Release

**iPhone** **armv7** **:**  
make P=iPhone/sim  
make P=iPhone/armv7 C=Release

( ):

make P=MacOS C=Release+NoRTTI Lib/MacOS-x86\_64/Release\_NoRTTI/libgfx.a

4 (-j4):

make P=PS3 C=Shipping+NoRTTI -j4 Bin/PS3/FxPlayer\_Shipping\_NoRTTI.elf

### 2.2.9.3 SCU

Scaleform (SCU)

. Lumping Unity . SCU  
. Scaleform SDK

SCU

. SCU

, Scaleform

SCU \_All.cpp

(SCU) #include

AS3 Obj Accessibility SCU

Src/GFx/AS3/Obj/Accessibility/AS3\_Obj\_Accessibility\_All.cpp

```
...  
#include "AS3_Obj_Accessibility_ISearchableText.cpp"  
#include "AS3_Obj_Accessibility_ISimpleTextSelection.cpp"  
#include "AS3_Obj_Accessibility_Accessibility.cpp"  
#include "AS3_Obj_Accessibility_AccessibilityImplementation.cpp"  
#include "AS3_Obj_Accessibility_AccessibilityProperties.cpp"
```

. , SCU

. Visual Studio SCU

SCU :

x AS2\_SCU

x AS3\_SCU

x AIR\_SCU

Non-SCU :

x AS2

x AS3

x AIR

non-SCU

make SCU

SCU

make

'SCU=0'

make P=PS3+snc C=Release+NoRTTI SCU=0

## 2.3 Windows Scaleform SDK

Scaleform on Windows® XP (32 bit, SP3), Vista (32/64 bit, SP2) Windows 7 (32/64 bit) . Visual Studio 2009 (Visual Studio 2010 ) .

### 2.3.1.1 Scaleform

Scaleform

Scaleform

## Scaleform SDK

```

\Program Files\Scaleform\GFx SDK 4.4\Lib\[Platform]\[MSVC Version]\

```

4

|                          |              |            |
|--------------------------|--------------|------------|
| Debug                    | Scaleform    | Debug libs |
| Runtime Library          | Debug (/MDd) |            |
| Debug Information Format | C7 (/Z7)     |            |
|                          |              |            |

| DebugOpt                 | Scaleform | Debug |
|--------------------------|-----------|-------|
| Runtime Library          | (/MDd)    |       |
| Debug Information Format | C7        | (/Z7) |
| Optimization             | (/Ox)     |       |
|                          |           |       |

| Release | Scaleform | Release |
|---------|-----------|---------|
|---------|-----------|---------|

|                 |       |
|-----------------|-------|
| Runtime Library | (/MD) |
| Optimization    | (/Ox) |
|                 |       |

| Shipping        | Scaleform | Shipping libs |
|-----------------|-----------|---------------|
| Runtime Library | (/MD)     |               |
| Optimization    | (/Ox)     |               |
|                 |           |               |

### 2.3.1.2 MSVC

Scaleform (SWFToTexture, TextureInSWF ) "Working  
 Directory" . C:\Program Files (x86)\Scaleform\GFx SDK  
 4.4\Bin\Data

1. Project .
2. Properties( ) .
3. Configuration All Configurations .
4. Debugging .
5. Working Directory .

## 2.4 Mac Scaleform SDK

Scaleform SDK                      Scaleform                      Cocoa      GL                      MacOS

.

### 2.4.1

Scaleform

x MacOS Snow Leopard(10.6)/Lion(10.7)

x Mac/iPhone/iPad      Xcode(      3.2.6      )

: PPC

.

### 2.4.2

MacOS      Scaleform

.

MacOS

.

x sf\_4.4\_eval\_macos\_i686.tar.bz2

sf\_4.4\_eval\_macos\_x86\_64.tar.bz2

src

x sf\_4.4\_src\_macos\_i686.tar.bz2

sf\_4.4\_src\_macos\_x86\_64.tar.bz2

(      )

src

x sf\_4.4\_lib\_macos\_i686.tar.bz2

sf\_4.4\_lib\_macos\_x86\_64.tar.bz2

x

o sf\_4.4\_{eval/lib/src}\_video\_macos\_i686.tar.bz2

### 2.4.3 MacOS X(10.6 10.7)

Scaleform MacOS SDK                      terminal.app                      Scaleform                      .

x86    x64    Scaleform(            )            :

\$ make P=MacOS

Bin/MacOS                      .

cpu    Scaleform(            )            :

\$ make P=MacOS/i686

\$ make P=MacOS/x86\_64

Zlib                      .

### 2.4.4

*Bin/MacOS-i686            Bin/MacOS-x86\_64            .*

Finder                      .

:    \$ *Bin/MacOS-i686/FxPlayer.app/Contents/MacOS/FxPlayer <swf file>*

Finder            SWF            Dock            ,            . HUD

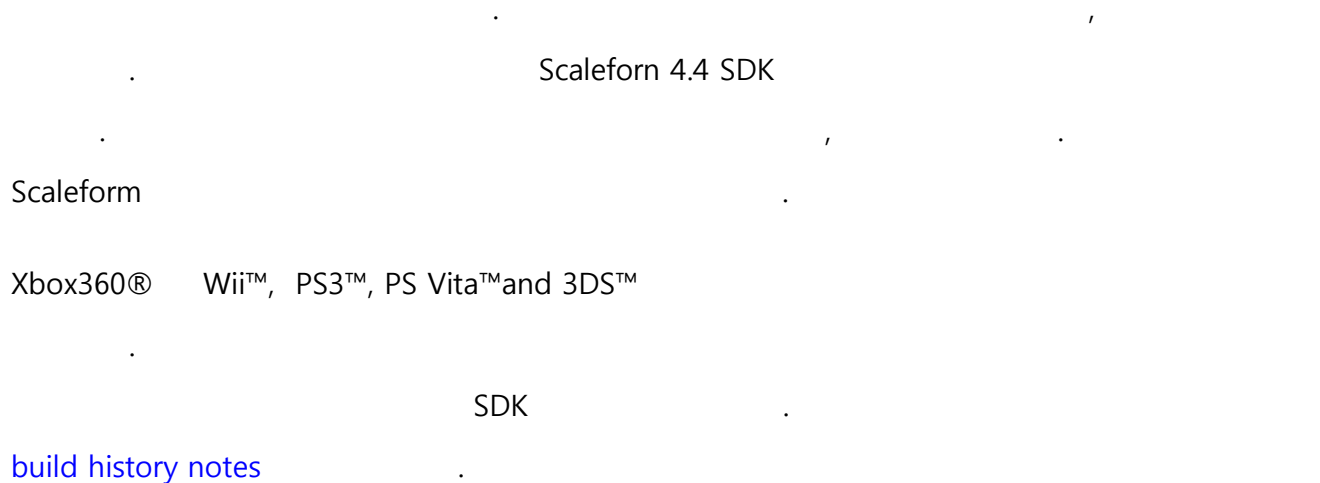
AS2    AS3            Finder(*Bin#Data#AS3#Kits#Menu            Bin#Data#AS3#Kits#HUD*)

.

(            ): *\$Bin/MacOS-i686/MenuDemo.app/Contents/MacOS/MenuDemo*



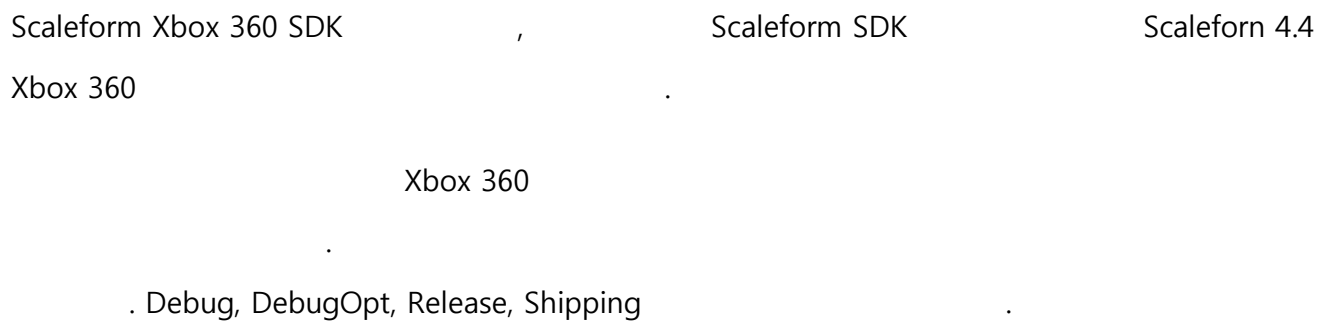
## 2.5 Scaleform SDK



### 2.5.1 Xbox 360 Scaleform SDK



#### 2.5.1.1



Scaleform  
 "Build" "Build Solution"  
 ( , Scaleform SWF ) Xbox  
 Xbox  
 Xbox " "  
 " " " "  
 . ( F5  
 .) XBox  
 Xbox . Xbox  
 " "  
 " "  
 . ( F5  
 .) Xbox . Scaleform 4.2 Xbox 360  
 Player, SWFTToTexture, TextureInSWF, PlayerTiny  
 Xbox , Xbox . Xbox  
 . Xbox  
 A . Xbox Player, SWFTToTexture, TextureInSWF,  
 PlayerTiny (GFx\_Xbox360,  
 GFxExpat, Sound)  
 Xbox 360® FxPlayer  
 x focus ( )  
 x A focus (Enter)  
 x Y  
 x X  
 x B

```

x      RB
x      LB
x      Start      /
x      Back      .

```

```

<back>      ,      FxPlayer      Flash
      .

```

```

x  A : Enter,
x  B  : Escape,
x  Start : ~,
x  LT : PgUp,
x  RT : PgDn,
x  LB: Home,
x  RB : End,
x  LTH : Insert
x  RTH : Delete

```

```

      FxPlayerXbox360.cpp      FXPLAYER_FILENAME      Flash      (SWF)
      .

```

```

:
#define FXPLAYER_FILENAME "Window.swf"
      "Window.swf"      focus      A      .

```

## 2.5.1.2

Scaleform (, Xbox360), Scaleform SDK .

Xbox360 Scaleform .

1. . Scaleform  
 . {GFX\_SDK}/Projects/Xbox360/Msvc90/SDK/GFx 4.4 Xbox 360 3rd Party.sln  
, (Release ) .

2. Open {GFX\_SDK}/Projects/Xbox360/Msvc90/SDK/GFx 4.4 Xbox 360 SDK.sln .

3. .

:

RTTI( ) C++ Xbox360

.

## 2.5.2 PS3 Scaleform SDK

PS3 Scaleform SDK SDK readme

.

### 2.5.2.1

Cell SDK 250 FMOD . GNU

3.81 . ( )

ProDG . PS3 SNC

.

libgfxrender\_ps3.a .

,

,

*Lib/PS3/Shaders.a*

*Lib/PS3/GLShaders.a*

•

1

•

**1**

V

```
letter>/... (Cygwin ) , CELL_SDK C:\ps3\cell
```

```
"export CELL_SDK := /cygdrive/c/ps3/cell".
```

### 3. Scaleform

```
$ make P=PS3
```

```
SNC :
```

```
$ make P=PS3+snc
```

```
 , "normal" PS3 snc ,
snc snc . (make P=PS3+snc
.)
```

### 4. \$(CELL\_SDK)FxPlayer/flash.swf SWF . SWF

```
Scaleform L1 R1 ScaleformPlayer .
```

### 5. "\$(CELL\_SDK)FxPlayer/sf\_license.txtsf\_license.txt"

```
.( cell .) RTC PRX .
```

```
cellSysmoduleLoadModule(CELL_SYSMODULE_RTC);
```

### 6. ProDG . FxPlayer

```
$(CELL_SDK)FxPlayer .
```

### 7. SWFToTexture RenderTexture ,

```
Bin/Data/AS2/Samples/SWFToTexture Bin/Data/AS2/Samples/RenderTexture
```

```
SWF $(CELL_SDK)FxPlayer .
```

## 2.5.2.4 Visual Studio

```
Make , Scaleform
```

```
.( PS3 )
```

Scaleform , PS3

*Projects/PS3/Msvc90/SDK* Scaleform 4.4 PS3 SDK.sln

*Projects/PS3/Msvc90/Demos*  
Scaleform 4.4 PS3 Demos.sln

## 2.5.2.5

"PS3" Scaleform

- focus ( )
- focus (Enter)
- 
- HUD
- 
- R1, L1 ,
- L2
- R2 , /
- Start /
- Select

<select> , FxPlayer FxPlayerAppBase  
ctor(PadKeyCommandMap ) Flash

```

: "Window.swf" focus (X)
. FxPlayer "Window.swf" /usr/local/cell/FxPlayer/flash.swf
. SWF/GFX L1/L2
.

```

### 2.5.2.6 PS3

```

, GCM GL PS3 .

```

libgcm:

```

: Src/Render/PS3/
.
Lib/PS3/Shaders.a , .
$ make P=PS3 Lib/PS3/Shaders.a

```

PSGL:

```

.
Lib/PS3/GLShaders.a , .
$ make P=PS3 Lib/PS3/GLShaders.a

```

```

GLShader.a SN . GNU "ppu-lv2-g++ -mno-sn-
ld" .

```

```

libcgm PSGL Scaleform player .

```

```

libgcm :
x PlayerGCM SWF
x PlayerTiny SWF
x SWFToTextureGCM SWF
x TextureInSWFGCM SWF

```



```

PSGL      :
    x PlayerGL                      SWF
    x PlayerTinyGL                  SWF
    x SWFToTextureGL                SWF
    x TextureInSWFGL                SWF

,

PSGL

```

libgcm .

```

PSGL  Apps/Samples/Common/OpenGLPS3App.*

SWFToTexture/TextureInSWF  Apps/Samples/Common/MathLib.*

.

```

```

#ifdef FXPLAYER_RENDER_GCM
    <gcm specific>
#else
    <psgl specific>
#endif

```

### 2.5.2.7

.(Release )

```

x Shipping .
x Release .
x Debug
x DebugOpt
x NoRTTI RTTI .
x NoThreads Scaleform .(SF_ENABLE_THREADS)

C=<config> . + .
, "make C=Release+NoThreads" .

```

### 2.5.2.8

Scaleform PRX . (resc, fs, gcm)

```
cellSysmoduleLoadModule(CELL_SYSMODULE_RESC);  
cellSysmoduleLoadModule(CELL_SYSMODULE_FS);  
cellSysmoduleLoadModule(CELL_SYSMODULE_GCM);
```

"Date" ( ) rtc .

```
cellSysmoduleLoadModule(CELL_SYSMODULE_RTC);
```

### 2.5.2.9

SF\_ENABLE\_THREADS . SF\_ThreadsPthead.cpp

## 2.5.3 PS Vita 8y Scaleform SDK

### 2.5.3.1

Vita SDK .

### 2.5.3.2

PS Vita    Scaleform SDK

PS Vita

x sf\_4.4\_psvita\_vc90\_eval.exe

src

x sf\_4.4\_psvita\_vc90\_src.exe

(       )

x sf\_4.4\_psvita\_vc90\_lib.exe

### 2.5.3.3 /

Scaleform PS Vita SDK

Scaleform 4.4 PS Vita Demos Visual Studio

. "Projects/PSVita/Msvc90/Demos/GFx 4.4 PSVitaDemos.sln"

FxPlayerMobile    FxPlayerTiny    Release, Debug, DebugOpt    Shipping

. PS Vita    Scaleform    Scaleform SDK

PS Vita    Scaleform

1. "Projects/PSVita/Msvc90/SDK/GFx 4.4 PSVita SDK.sln"

2. Build Solution

make

Vita

2.2.9 'make

Scaleform

### 2.5.3.4

FxPlayerTiny

Scaleform

FxPlayerMobile

HUD

(SWF ) Vita "File Serving Directory" "FxPlayer"

FxPlayerMobile "flash.swf" SWF

"flash.swf"

### 2.5.3.5 F V .Ú:Z

2.10

## 2.5.4 Wii Scaleform SDK

Wii Scaleform SDK read me (readme\_wii.txt, readme\_make.txt)

### 2.5.4.1

Bin/wii Codewarrior ndrun . Codewarrior

, . F5

. ( .)

DVD

. RVL\_SDK/dvddata/FxPlayer

dvddata/FxPlayer/gfxlicense/txt

Player flash.swf (SWF GFX )

PlayerTiny Window.swf (SWF GFX )

TextureInSWF     *Bin/Data/AS2/Samples/RenderTexture*  
                              *dvddata/FxPlayer* .

- and + (dvddata/FxPlayer SWF GFX )  
C  
Z .

```
<Z> , FxPlayer FxPlayerAppBase
ctor(PadKeyCommandMap ) Flash
```

### 2.5.4.2

|          |   |
|----------|---|
| Shipping | . |
| Release  | . |

Debug

DebugOpt

+NoRTTI

RTTI

+Sdata0

No small data section (-sdata 0 -sdata2 0)

+NoThreads

Scaleform

(SF\_ENABLE\_THREADS,

.)

.Wii

+NoRTTI

RTTI

+ \_

Codewarrior  
projects"

, Preference

IDE Extras group

"Use text-based

XML

. Codewarrior

Make

libgfx\_video.a

C/C++

"#define GFX\_ENABLE\_VIDEO"

Scaleform

. (Cygwin

. PS3

2.3.2.3

Scaleform

Cygwin

.)

```

, Wii SDK          SDK          Makeconfig          .(
.)

```

```

WII_CWDIR := "/cygdrive/c/Program Files/Freescale/CW for Wii v1.0"
WII_SDKDIR := /cygdrive/c/RVL_SDK
WII_NDEVDIR := /cygdrive/c/NDEV

```

```

Makeconfig          ,          ,          .
/cygdrive/<drive letter>/...          .
.

```

```

cygwin          "make P=wii C=DebugOpt+NoRTTI"          .
DebugOpt+NoRTTI          .
.          C          . Makeconfig
RVL_NDEV          .

```

## 2.5.5 Nintendo 3DS 8ÿ Scaleform SDK

### 2.5.5.1

```

CTR SDK          .

```

### 2.5.5.2

```

3DS          Scaleform SDK          .
3DS          .
.
x sf_4.4_eval_3ds.tar.bz2
src          .
x sf_4.4_src_3ds.tar.bz2

```

( )  
sf\_4.4\_lib\_3ds.tar.bz2

### 2.5.6.3

Scaleform 3DS SDK                      Cygwin                      Scaleform                      .

Scaleform( ) :

\$ make P=3DS

Bin/3DS .

2.2.9 'make      Scaleform      ' .

### 2.5.6.4

FxPlayerMobile / FxPlayerTiny                      Projects/3DS/Makefile, STARTUP\_FLASH

FxPlayer\_CONTENT                      Flash .

.      SWF                      Obj/3DS/rom\_FxPlayerMobile/FxPlayer                      cci

.

CTR .

### 2.5.6.5

2.10 .



## 2.6 Scaleform SDK

Scaleform

Scaleform

iOS

### 2.6.1 Android 8y Scaleform SDK

#### 2.6.1.1

Scaleform

- x [Android SDK](#)
- x [Android NDK](#)
- x Android Platform API

#### 2.6.1.2

Android Scaleform SDK

Android

x sf\_4.4\_eval\_android.tar.gz

src

x sf\_4.4\_src\_android.tar.gz

( )

x sf\_4.4\_lib\_android.tar.gz

#### 2.6.1.3 /

# 1. Scaleform Makeconfig

```
export JAVA_HOME := <JDK >
ANDROID_SDK := <SDK >
ANDROID_NDK := <NDK >
export PATH := $(JAVA_HOME)/bin:$(ANDROID_SDK)/tools:$(ANDROID_SDK)/platform-
tools
```

```
Makeconfig (slash) UNIX
.msys /< >/... , Cygwin /cygdrive/<
>/... , Android_SDK C:\Android\SDK
```

```
"export ANDROID_SDK := /cygdrive/c/Android/SDK".
```

# 2. Scaleform

```
$ make P=Android
```

# 3. Bin/Android APK . APK

# 4. SWF /sdcard/flash.swf . SWF GFx HUD Scaleform Player .

# 5. "/sdcard/sf\_license.txtsf\_license.txt" .

```
flash.swf
"/sdcard"
```

## 2.6.1.4

```
Scaleform                                Scaleform SDK
Android  Scaleform                      . Make
```

## 2.6.2 iOS Scaleform SDK

### 2.6.2.1

```
Scaleform                                .
```

- x Mac OS X Snow Leopard
- x Xcode 3.2.5, Xcode 4
- x SDK

### 2.6.2.2

```
iOS  Scaleform SDK                      iOS
```

- x sf\_4.4\_eval\_ios.tar.bz2

```
src
```

- x sf\_4.4\_src\_ios.tar.bz2

```
( )
```

- x sf\_4.4\_lib\_ios.tar.bz2

### 2.6.2.3 /

```
Scaleform iOS                          "Project\iPhone\Xcode\Gfx 4. 3iPhone SDK" Scaleform
4.4 iPhone Xcode                      .
```

iOS . Xcode  
 .  
 Debug, DebugOpt, Release Shipping .  
 Scaleform . Xcode "Build  
 and Run" .  
 ( GFX SWF )  
 iOS .  
 iOS "flash.swf" SWF .  
 SWF iTunes  
 iTunes Apps "File  
 Sharing" , Apps FxPlayerMobile SWF "FxPlayerMobile  
 Documents" "Add" SWF .  
 .

#### 2.6.2.4

Scaleform Scaleform SDK . iOS  
 Scaleform .  
 1. Scaleform iOS . lipo (   
 Release ) make ( :  
 make P=iPhone/armv7 C=Release

#### 2.6.3 (FAQ)

Scaleform SDK for Mobiles ,



## 2.7 CS3 Scaleform

Scaleform SWF Scaleform player .

.

1. ( → → Adobe Extension Manager CS3) C:/Program Files/Adobe/Adobe Extension Manager.

Finder>Applications>Adobe Extension Manager

CS3>Extension Manager .

2. ,

.

3. .  
: C:/Program Files/Scaleform/GFx SDK 4.4/Resources/Tools/  
: scaleform\_gfx\_4.4\_macos/Resources/Tools/

4. Scaleform Extensions.mxp

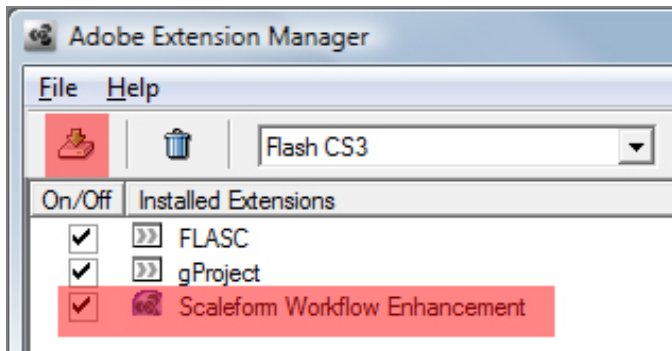
**1: Scaleform Extensions.mxp**

**(Windows)**

**2: Scaleform Extensions.mxp (Mac)**

5. *Accept*

6. OK



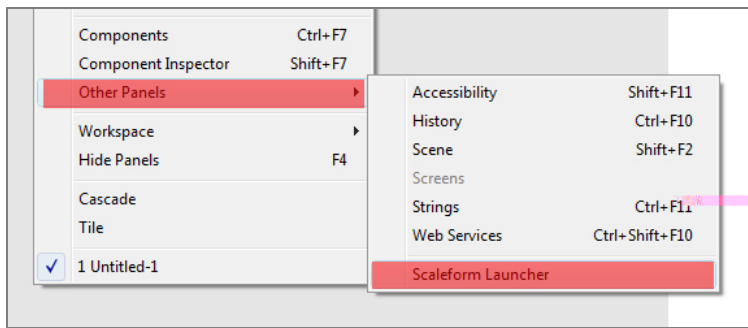
**3: CS3 (Windows)**

**4 CS3 (Mac)**

7.

8. Flash , Flash ( 2.0 or 3.0)

9. Flash *Other Panels* Scaleform Launcher .



**5: Other panels**

**Scaleform Launcher (Windows)**

**6: Other panels**

**Scaleform Launcher (Mac)**

## 2.8 CS4, CS5 Scaleform

Scaleform SWF Scaleform player .

.

1. Help Manage Extensions .

: Flash

. 2 .

2. Install

3.

: C:/Program Files/Scaleform/GFx SDK 4.4/Resources/Tools/

: *scaleform\_gfx\_4.4\_macos/Resources/Tools/*



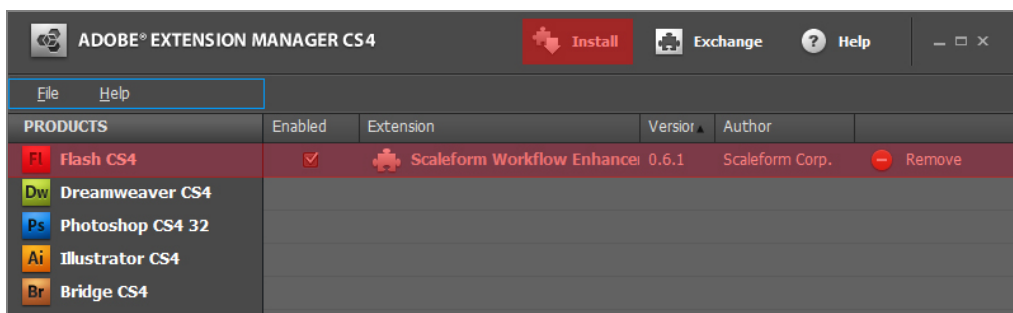
4. *Scaleform Extensions.mxp*

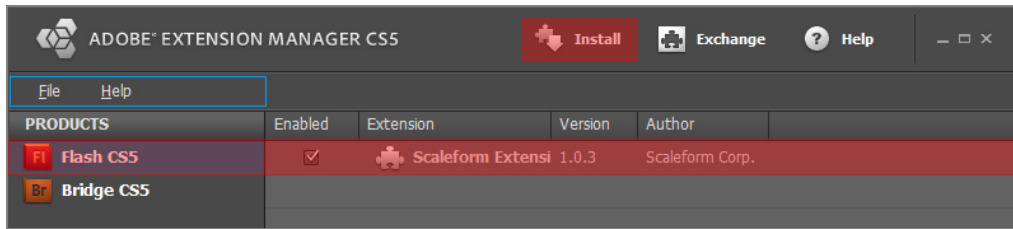
**7: Scaleform Extensions.mxp (Windows)**

**8: Scaleform Extensions.mxp (Mac)**

5. *Accept*

6. *OK* CS4 or CS5



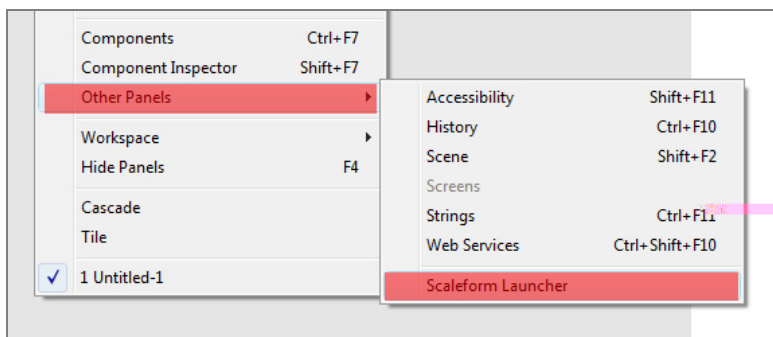


## 9: CS4(top) or CS5 (bottom)

7.

8. Flash

9. Flash     *Other Panels*                      Scaleform Launcher



## 10: Scaleform (Windows)

## 11: Scaleform (Mac)



. [+]

Add New Profile

.

**13: [+]**

:

NO

.

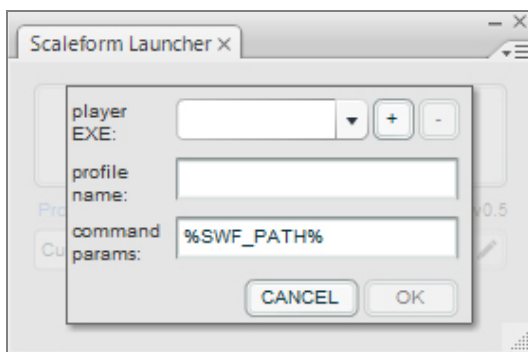
.

**14:**

2.

[+]

Scaleform



**15: Add new profile**

.

3.

GFxMediaPlayer

.

.

,

Scaleform

.

GFxMediaPlayer

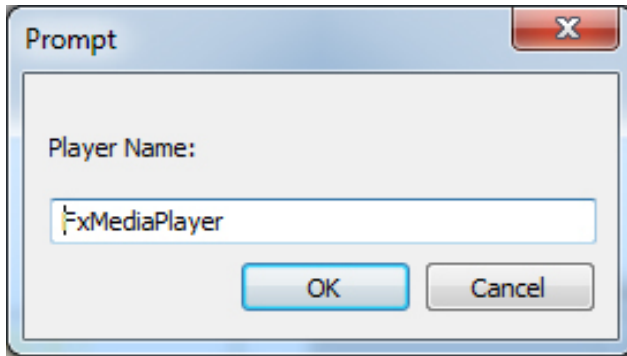
. C:/Program Files/Scaleform/GFx SDK 4.4/Bin

## **16: GfxMediaPlayer (Windows)**

scaleform\_gfx\_4.4\_macos/Bin/MacOS/FxPlayer\_DebugOpt.app/Contents/MacOS  
GfxPlayer\_DebugOpt .

## **17: GfxPlayer\_DebugOpt (Mac)**

4. .

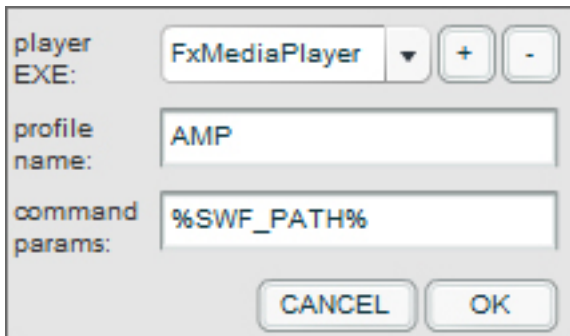


18:

19:

5. OK .
6. 'DebugOpt' Add Profile .  
AMP .
7. .

'%SWF\_PATH%' , .  
: '%SWF\_PATH% -nh'  
HUD SWF GFx .



20:

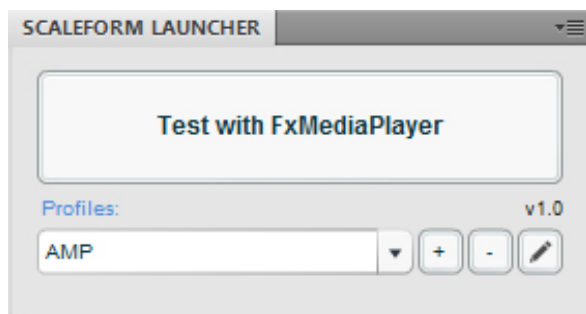
AMP

(windows)

**21: GfxPlayer\_DebugOpt (Mac)**

8. OK .

9. Profiles Scaleform .



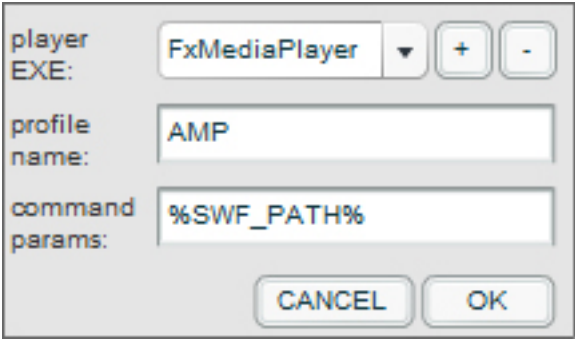
**22: AMP Scaleform**

**23: Debug\_Opt Scaleform**

1. .

2. Edit

3. , , .
4. OK



24: / (Windows)

25: / (Mac)

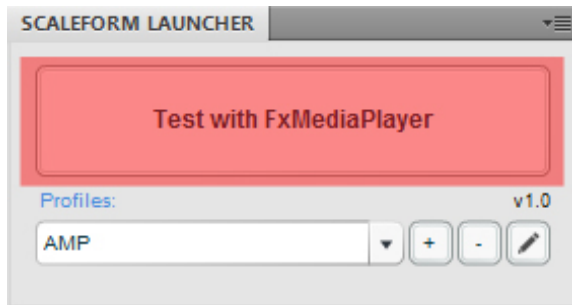
1. Profile
2. [-] .
3. OK . Cancel

**SWF**

1. SWF .
2. .
3. Test with [ ] .



4. Scaleform player movie .



**26: Test with GfxMediaPlayer (Windows)**

**27: Test with GfxPlayer\_DebugOpt (Mac)**

Flash Scaleform Player .

1. *Edit* .

2. *Keyboard Shortcuts* .

3. Adobe Standard Current . (Keyboard Shortcuts )

4. *Duplicate Set* .

5. ( ) .

6. *OK* .
7. "*Commands* Drawing Menu Commands" .
8. *Commands* + .
9. Scaleform Player Publish .
10. + .
11. Press (Ctrl+Enter) Ctrl+Enter .  
Flash .
12. *Change* .
13. , *Reassign* .
14. *Commands* Control .
15. Test Movie .. Flash publish .
16. + .
17. Press (Ctrl+\ (backslash)) .
18. *Change* .
19. *OK* Keyboard Shortcuts .

## 2.10

### 2.10.1

Scaleform 4.4

*sf\_license.txt*

*. sf\_license.txt*

3AAAA1BB23B8ZCCCC4CCEOFSJCZ08DDDDDEF

*sf\_license.txt*

. Scaleform 4.4  
SDK

*sf\_license.txt*

*sf\_license.txt*

*sf\_license.txt*

Scaleform 4.4

The Scaleform trial period expired on 1(m) 1(d) 2008, please contact Scaleform for an extension. Error: 106

*sf\_license.txt*

Failed to open the Scaleform license file, please make sure sf\_license.txt exists in the working directory. Error: 101

## Windows / Mac® / Linux®

- 1.
2. GFXSDK

3. GFXSDK , HOME

### **Xbox 360®**

1. XDK Debug Drive - D:/\${SolutionName}/sf\_license.txt

.

### **“PSP®” (PlayStation®Portable)”**

1. :  
Linux - /usr/local/psp/FxPlayer/sf\_license.txt  
Windows - C:/usr/local/psp/FxPlayer/sf\_license.txt

### **“PLAYSTATION®3”(PS3™)**

1. /app\_home/sf\_license.txt
2. /host\_root/usr/local/cell/FxPlayer/sf\_license.txt
3. /dev\_bdvd/PS3\_GAME/USRDIR/sf\_license.txt

### **PLAYSTATION®Vita”(PS Vita™)**

1. "FxPlayer" sf\_license.txt

.

### **3DS™**

1. sf\_license.txt Lib/3DS/target .

### **Wii™**

1. dvddata/FxPlayer/sf\_license.txt

## iOS

1. sf\_license.txt Xcode Xcode ( :  
Scaleform.app) .

## Android

1. sf\_license.txt / "sdcard"  
( root  
).).

## 2.11

Scaleform

: Scaleform

<http://gameware.autodesk.com/scaleform>

<http://gameware.autodesk.com/scaleform/developer/>

Scaleform , , , ,

, FAQ .

: .

, e-mail

. e-mail ,

, , , ( ) . ,

Scaleform

: Scaleform

Scaleform

Scaleform

e-mail

## 2.12

Scaleform

[gameware@autodesk.com](mailto:gameware@autodesk.com)

Scaleform

### 3

Scaleform 4.4 . Scaleform 4.3  
 , Scaleform 4.4 , Scaleform 4.3 ,  
Scaleform 4.4 Flash AS 2.0 .  
 , , , ,  
 , , , ,  
Scaleform 4.4 , Doc  
 .  
 .  
1. SDK  
2. : → → Scaleform → GfX SDK 4.4 → Documentation

#### 3.1 *Flash*

Flash AS2 . Flash  
Scaleform IGDA (International Game Developers Association) Flash SIG (Special  
Interest Group) [http://www.igda.org/wiki/Flash\\_SIG](http://www.igda.org/wiki/Flash_SIG) .  
Flash , IGDA SIG  
 .  
Flash  
x [Lynda.com](http://www.lynda.com)  
x [Essential ActionScript 2.0 by Collin Mooch](#)  
x [ActionScript: The Definitive Guide by Collin Mooch](#)



3.2

Scaleform 4.4

Scaleform 4.4

- x Getting Started with CLIK : CLIK Flash, CLIK,
- x Getting Started with CLIK Buttons : CLIK Getting Started with CLIK
- x Getting Started with Video : , USM video exporting, Video Flash Video : Scaleform Video
- x Getting Started with IME : IME

3.3

Scaleform 4.4 Flash

4p.

- x Integration Tutorial: Scaleform 4.4 3D.328 1080x2160 53B 0121 0301 1TW2348

- x ActionScript Extensions: Scaleform 4.4
- x Font Overview Configuration: Scaleform 4.4
- API Scaleform C++
- x Scale9Grid: Scaleform 4.4
- Scale9Grid
- x XML Overview: Scaleform 4.4 XML
- x Flash Support Overview: Scaleform 4.4 Flash
- (XLS)
- x Memory System Overview: Scaleform 4.4 AMP,
- IME Configuration: Scaleform 4.4 IME
- IME

### 3.4

- , UE3
- SDK
- [gameware@autodesk.com](mailto:gameware@autodesk.com)
- x IME Xbox 360 Configuration: Xbox 360 IME
- x IME PS3 Configuration: PS3 IME
- x Unreal Engine 3 Workflow: Flash 3

x Unreal Engine 3 Integration: Scaleform 4.4 UE3  
 . Scaleform 4.4 Kismet, ,  
 UE3 .

x README -  
 readme . readme SDK .

### 3.5

Scaleform 4.4 . Scaleform  
 . Demos

"Downloads"

<http://gameware.autodesk.com/scaleform/developer/?action=demo> .

Video, , Flash Scaleform 4.4

x *Scaleform Player Tiny*. Scaleform Player  
 . Scaleform SDK

( ): C:/Program Files (x86)/Scaleform/GFx SDK  
 4.4/Apps/Samples/GFxPlayerTiny

x *Scaleform Player SWF to Texture*: Flash 3D  
 . 3D .  
 ( : ).

(PC): C:/Program Files/Scaleform/GFx SDK 4.4/Bin/Win32/Msvc90/  
 GFxPlayerSWFToTexture/SWFToTexture\_D3D9\_Release\_Static.exe  
 ( ): C:/Program Files (x86)/Scaleform/GFx SDK  
 4.4/Apps/Samples/RenderTexture

x *Scaleform Player Textures in SWF.* Flash 3D  
 . 3D tween Flash .

(PC): *C:/Program Files/Scaleform/GFx SDK 4.4/ Bin/Win32/Msvc90/  
 GFxPlayerTextureInSWF/TextureInSWF\_D3D9\_Release\_Static.exe*  
 ( ): *C:/Program Files (x86)/Scaleform/GFx SDK  
 4.4/Apps/Samples/RenderTexture*  
 Scaleform SDK *Bin/<platform>/ directory*

## 3.6

Scaleform 4.4 SDK Scaleform  
 : <http://gameware.autodesk.com/scaleform/developer/?action=doc>

Scaleform 'Getting Started' Scale SDK

<http://gameware.autodesk.com/scaleform/developer/?action=vid>

## 4

## Scaleform 4.4

Player

### Scaleform 4.4

Scaleform 4.4

Scaleform

Scaleform 4.4

Scaleform 4.4

Scaleform 4.4

UI /

1

UI 1

Flash

Flash

). FAQ

• <http://gameware.autodesk.com/developer/forums/>

## 4.1

### 4.1.1 UI

Scaleform

DB

## 2D Flash

Scaleform 4.4

DB (

) Flash

Flash

( : , , HP ).

## 4.1.2 UI

Flash , AS2  
CLIK ,

## 4.1.3 UI

CS3 CS4, Flash ( CLIK )  
UI ,

Scaleform 4.4 Flash

## 4.2

Scaleform 4.4 . Scaleform 4.4  
SWF Scaleform Scaleform  
4.4 player SWF GFX Scaleform

Windows      Scaleform 4.4 SDK      "Scaleform SDK Browser"

                 . SDK Browser      'Programs'      Scaleform

Player      .

→      → Scaleform → GfX SDK 4.4 → GfX Players

*C:/Program Files/Scaleform/GfX SDK 4.4/Bin/Win32/Msvc80/GfXPlayer*

*C:/Program Files/Scaleform/GfX SDK 4.4/Bin/Win32/Msvc90/GfXPlayer*

SDK Browser      Scaleform GfX Player      .

C:/Program Files/Scaleform/GfX SDK 4.4/Bin/Data/AS2/Samples/3DDemo/ .      3D

SWF      .

Scaleform SDK 4.3

Bin/<platform>      .      Finder(      )

SWF      , Scaleform 4.4      Bin      3DDemo      SWF

SWF      .      SWF

- Bin/i386-linux/FxPlayer <swf file>

OS - Bin/MacOs/FxPlayer.app/Contents/MacOS/FxPlayer <swf file>

: Scaleform SDK 4.4      */Bin/Data/AS2/Samples*

## 4.3

Scaleform 4.4 AS 2.0, 3.0 . Flash CS3, CS4 CS5

. , Flash AS

2.0 AS3.0 . Flash File → New → Flash File

(ActionScript 2.0 ActionScript 3.0) .

Scaleform 4.4 AS 2.0, 3.0 .

*sf\_4.4\_flash\_support.pdf* AS 2.0 .